Native qualification diff adapter proposal

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The Rust library implements exact diff parsing together with identity, terminal-state, mount and security prerequisites. All 58 fixed native controls pass: nine positive cases and 49 negative cases. The component is ready for independent source review. A full native supervisor/callsite is not yet implemented.

Implemented contract

The public functions accept raw bytes and structured inspection/expected-identity data. Qualification is restricted to Q1–Q3 and requires the full container/image IDs, nonce-bound name/label, user/entrypoint, correct stopped exit state, read-only rootfs, no network, resource limits, privilege/namespace/device restrictions, exact read-only fixture/gate mounts and approved tmpfs. Only then are successful transport and the exact three added paths accepted.

Malformed identity, mount, byte framing or transport conditions return errors. Accepted results retain null writer attribution and a false no-writes claim. The library does not collect daemon evidence or authenticate an executed container; those remain caller responsibilities.

Controls and preserved source

Controls test all three valid cases and six diff orderings, plus wrong identities, user/entrypoint, running/wrong exit state, security/resource restrictions, wrong/writable/duplicate/missing mounts, transport failure and malformed bytes. A second replay matches; source hashes remain unchanged.

The v5 Python supervisor, published parser and failed receipt are unchanged reference evidence. An initial unexecuted Python adapter draft was stopped when the native-only repository rule was identified; it is retained as rejected provenance. The deliverable is Rust and neither executes nor extends the Python path.

Precise integration boundary

The proposed patch adds the full Rust library and fixed-control binary. It does not modify a supervisor. A native supervisor must still collect trusted identity/mount evidence, enforce command/output limits and exact-ID cleanup, preserve Q0–Q4 behavior and compose this adapter at the diff boundary. Its actual callsite needs independent fixed controls and review.

The binding template leaves review, supervisor, image/binary/fixture/gate and admission fields unfilled. Limits remain 300 seconds total,30 seconds cleanup reserve and20 seconds per active case if subsequently admitted. This library has no clock or runtime authority.

`REPORT.md` maps all eight amendment requirements to implemented and pending work; `binding-template.json`, `controls.json`, full source and patch provide reviewable artifacts. No Docker, qualification subprocess, supervisor, cloud or scientific execution occurred. The producer packet awaits independent review.